Cars and gears

13a

In a driving performance test, a car was put through its paces in order to find out the top speeds for the various gears. In first gear, it took three seconds to reach a speed of fifty kilometres per hour from standstill. Following a quick change into second gear, in another four seconds a speed of eighty kilometres per hour was reached. Third gear allowed a top speed of one hundred and ten kilometres per hour in a further six seconds. It took eleven seconds in fourth gear to reach one hundred and forty kilometres per hour. To reach the top speed of one hundred and sixty kilometres per hour required thirteen seconds in fifth gear. Finally, it took thirteen seconds of hard braking to bring the car to a complete stop.

1 What is the top speed of the car? _____

2 How long did it take the car to reach its top speed? _____

3 Which gear has shown the biggest speed change? _____

4 Which gear enables the fastest changes in speed? (Gives the greatest acceleration?) _____

5 Make a table of the results using these headings: Gear, Initial speed, Final speed, Speed change, Time taken, Total time. (Don't forget to include the units in the headings!)

Gear					

Acceleration is defined as the rate of change in velocity over time. (Velocity has the same magnitude as speed, but has a direction as well.) In our case, the speeds have been measured in kilometres per hour and the times in seconds. Therefore the unit for acceleration here is kilometres per hour per second, or km/h/s. We can find the change in speed by subtracting the initial (i.e. first) speed from the final (i.e. second) speed.

6 **a** Write a formula to find the change in speed.

_____

b Scientists usually write the letter **a** for acceleration, **t** for time, and **v** for speed. Initial speed is $\mathbf{v_1}$ and final speed $\mathbf{v_2}$. Rewrite your formula for the change in speed using abbreviations.

_____

7 State a unit for speed _____ and a unit for acceleration _____.

8 **a** To calculate acceleration, the change of speed is divided by the time it took to change that speed. Write a formula to calculate acceleration.

_____

b Rewrite your formula for acceleration using abbreviations. _____

9 Calculate the acceleration for each gear. (Don't forget to show your working out!)

Gear	Speed change	Speed change/Time taken	Acceleration
First			
Second			
Third			
Fourth			
Fifth			

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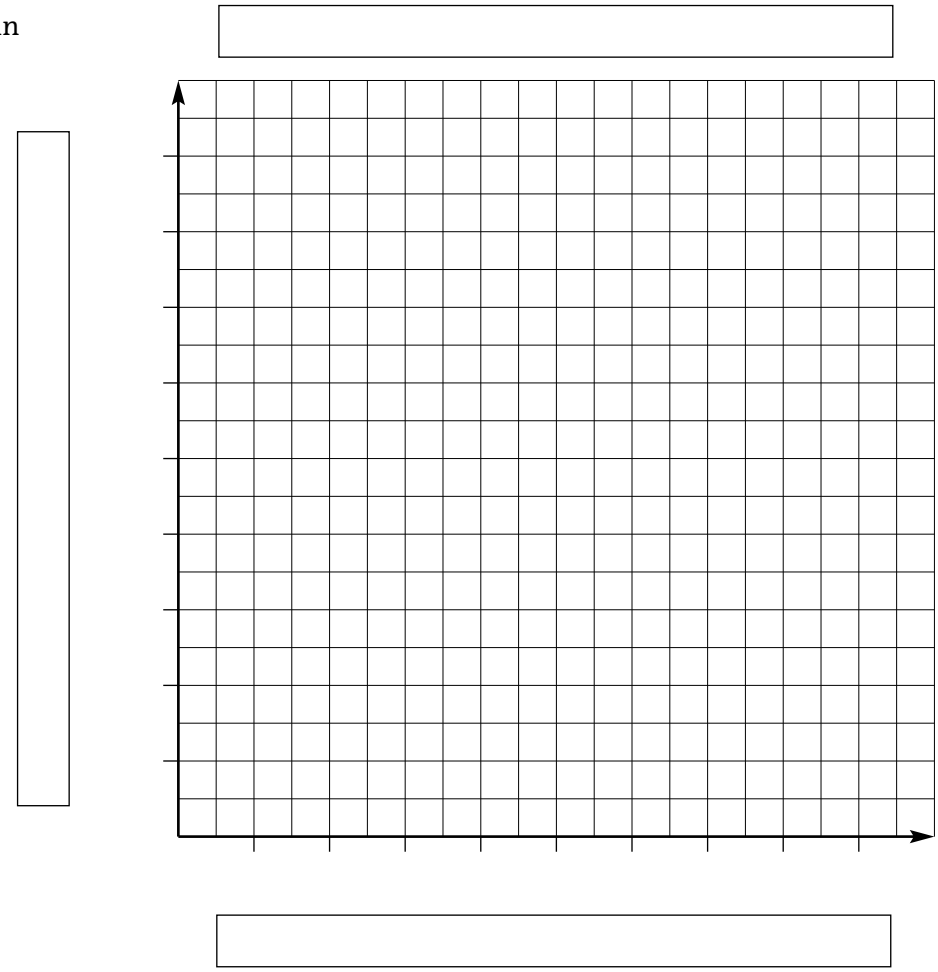
10 Record the speed results in a speed–time graph. (Hint: Time goes on the horizontal axis.)



11 **a** State how you can tell from the graph which gear produces the greatest acceleration.

b State how you can tell from the graph which gear produces the smallest acceleration.

c Explain how you could calculate the acceleration produced by a particular gear from the graph.

12 **a** Draw a line on the graph to represent the average acceleration of the car as it travelled from standstill to top speed.

b Calculate the average acceleration of the car. _____

13 A positive acceleration occurs when the car is becoming faster – the speed increases as time passes, so the final speed is higher than the initial speed. A negative acceleration (or deceleration) happens when the car is slowing down – the speed decreases as time increases, so the final speed is lower than the initial speed.

a State how you could identify a positive acceleration on a speed–time graph.

b State how you could identify a negative acceleration on a speed–time graph.

c Describe what no acceleration (i.e. a constant speed) looks like on a speed–time graph.

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